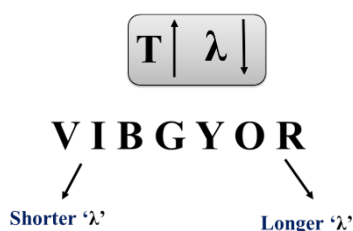
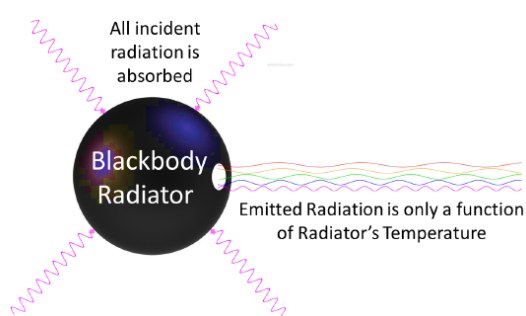


22PH202 – ELECTROMAGNETISM AND MODERN PHYSICS

Unit-IV: MODERN PHYSICS_LM5

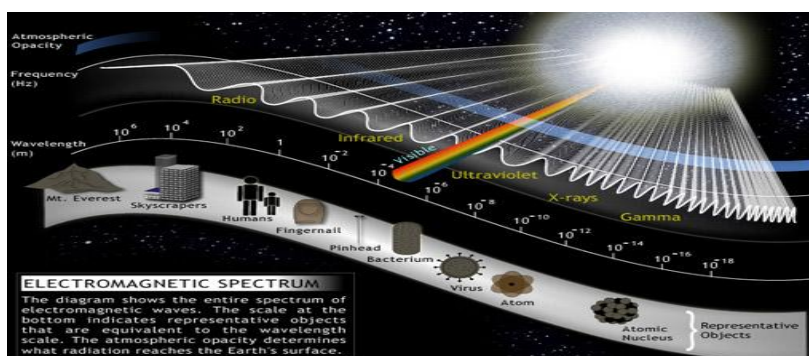
Black body radiation & Origin of Quantum Mechanics

- ❖ A surface that absorbs all radiation incident upon it would look **black**
- ❖ A **perfect black body** is an **ideal absorber & Ideal emitter** of thermal energy
- ❖ Ex: **Sun&Stars** are Ideal emitter



Platinum wire

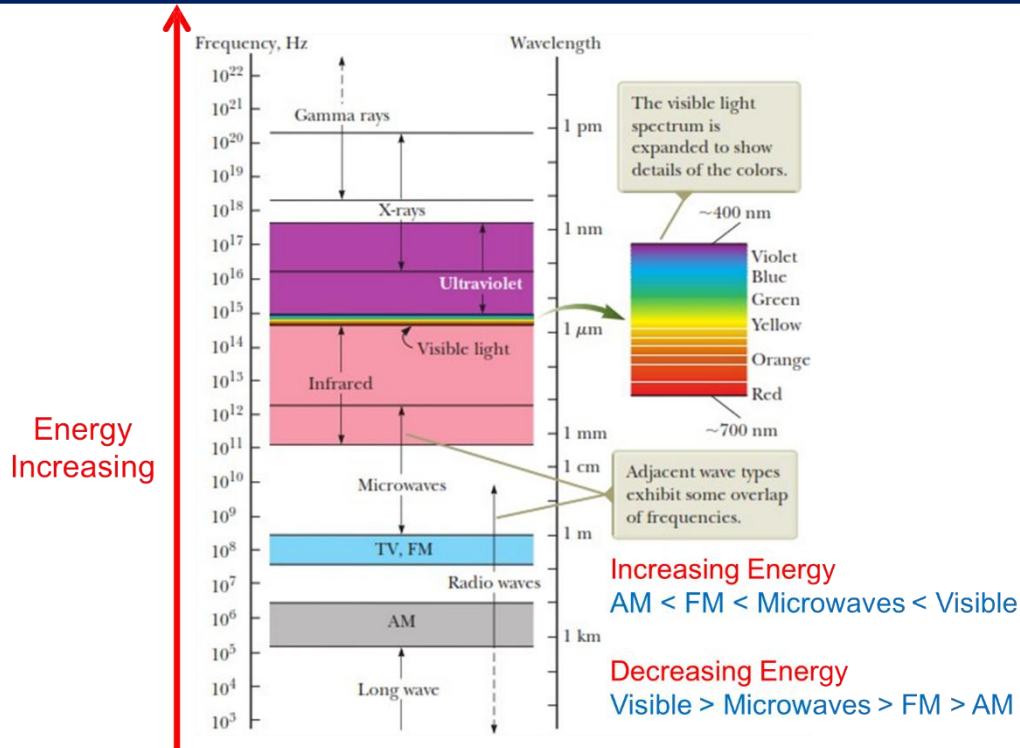
Temp	Color
500 °C	Dull Red
900 °C	Dark Red
1100 °C	Orange
1300 °C	Yellow
1600 °C	White



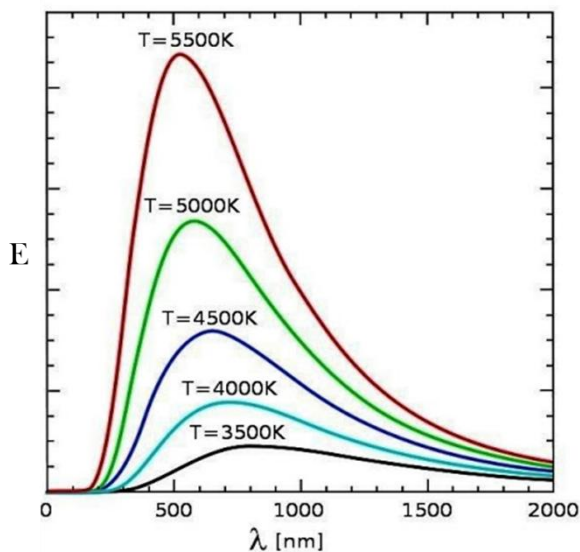
While standing outdoors one evening, you are exposed to the following four types of electromagnetic radiation:

- Yellow light from a sodium street lamp
- Radio waves from an AM radio station
- Radio waves from FM radio station
- Micro waves from an antenna of communication system

Rank these types of waves in terms of photon energy from the highest to the lowest.



Energy distribution Curve



- Energy distribution is not **Uniform**
- If wavelength increases, the energy emitted from black body increases and reaches **maximum at one wavelength** and decrease beyond maximum wavelength
- As temperature increases, the maximum wavelength shift towards **shorter wavelength**.
- Area under the curve is total **energy emitted by black body** per unit area per unit time.

Classical Theory

Kirchoff's Law (1859)

At any temperature, the ratio of emissive to absorptive power of a substance is equal to emissive power of the black body

$$E_{\lambda} = \frac{e_{\lambda}}{a_{\lambda}}$$

Wien's Law (1893)

$$\lambda_{\max} T = \text{Const}$$

$$\lambda_{\max} T = b$$

$$b = 2.9 \times 10^{-3} \text{ mK}$$

$$E_{\lambda} = \frac{A}{\lambda^5} d\lambda$$

Stefan's Law (1879)

$$E \propto T^4$$

$$E = \sigma T^4$$

$$\sigma = 5.6 \times 10^{-8} \text{ Wm}^{-2}\text{K}^{-4}$$

Stefan Boltzmann constant

Raleigh-Jeans Law (1900)

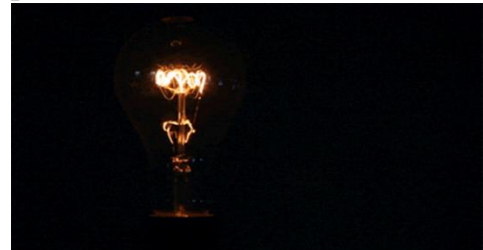
$$E_{\lambda} = \frac{8\pi K T}{\lambda^4} d\lambda$$

Stefan's Law or Stefan's Boltzmann's Law

The goal of this experiment is to investigate the relationship between and for the tungsten filament in an ordinary lamp to see how close it behaves like a black body radiator. The emissivity of tungsten is not quite constant but decreases with increase in wavelength and increases in temperature.

The total power of the **emitted radiation increases with temperature**, where we introduced Stefan's law:

$$P = \sigma A e T^4$$

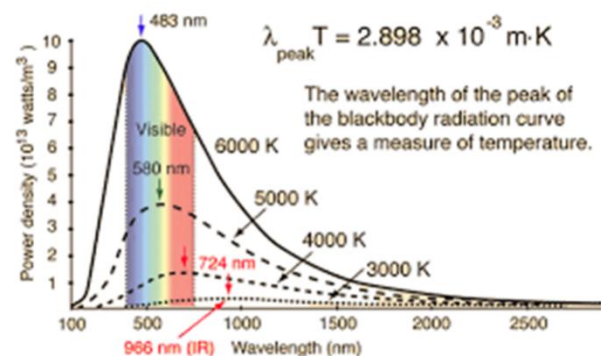
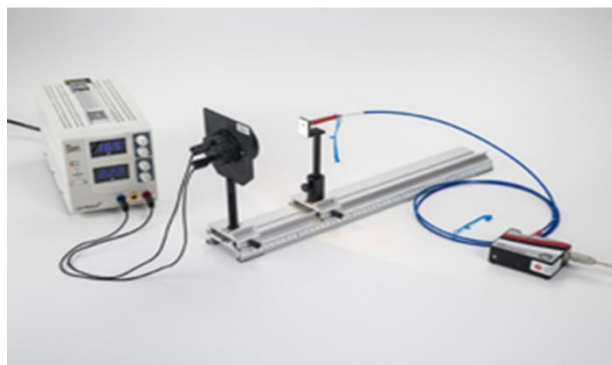


Where, P - the power in watts radiated at all wavelengths from the surface of an object, $\sigma = 5.6696 \times 10^{-8} \text{ W/m}^2 \text{ K}^4$ is the Stefan-Boltzmann constant, A- the surface area of the object in square meters, e is the emissivity of the surface and T is the surface temperature in kelvins.

For a black body, the emissivity is $e = 1$ exactly.

Wein's displacement law

The black body radiation curve for different temperature peaks at a wavelength inversely proportional to the temperature.



$$\lambda_{\text{max}} = \frac{c}{T} = \frac{2.898 \times 10^{-3}}{T}$$

Rayleigh-Jeans Law

- An early classical attempt to explain blackbody radiation was the **Rayleigh-Jeans law**.
- Energy distribution is **directly proportional to Absolute temp.** and **inversely proportional to fourth power of wavelength of radiation of hot body.**

$$E_{\lambda} = \frac{8\pi K T}{\lambda^4}$$

- At long wavelengths, the law matched experimental **results fairly well**.

Failures of Classical Theory

Limitations

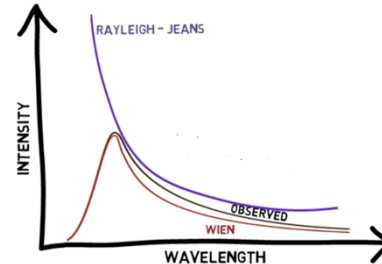
Rayleigh-Jean's Law holds good for **longer wavelength**. Also, it predicts that energy output diverges towards **infinity**.

Wien's Law holds good for shorter wavelength.



MAX PLANCK (1900)

Turned UV catastrophe into birth of **Quantum Physics**



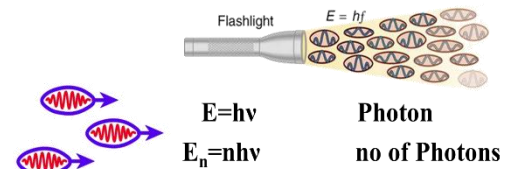
Assumption of Quantum theory of radiation (Planck)

A black body radiation chamber is filled with no. of oscillators; each oscillator vibrates with its own frequency.

- ❖ Each Oscillator has energy given by integral multiple of $h\nu$

$$E = nh\nu \text{ where } n=1,2,3\dots$$

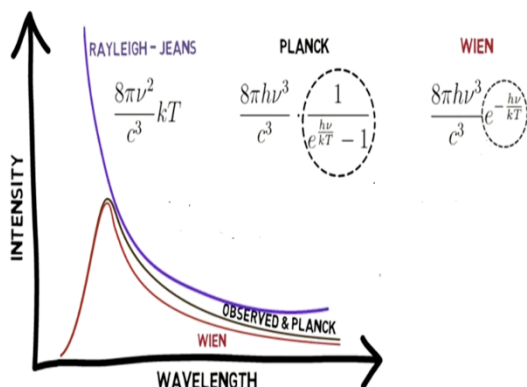
- ❖ The oscillator emits or absorb energy when making **transition from one quantum state to another**.



$$E_n = nhc/\lambda \quad E \propto \nu \text{ or } 1/\lambda$$

$$E_\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{hc/\lambda KT} - 1} d\lambda$$

Energy distribution curve for Black body radiation



Reduction of Planck's Law to Classical laws

$$E_\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{hc/\lambda KT} - 1} d\lambda$$

At shorter wavelength

$$\frac{1}{e^{hc/\lambda KT}} \gg 1$$

$$\frac{1}{e^{hc/\lambda KT} - 1} = \frac{1}{e^{hc/\lambda KT}}$$

$$E_\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{hc/\lambda KT}} d\lambda$$

Wien's Law

At longer wavelength

Expanding $e^{hc/\lambda KT}$ as power series

$$e^{hc/\lambda KT} = 1 + \frac{hc}{\lambda KT}$$

$$e^{hc/\lambda KT} - 1 = 1 + \frac{hc}{\lambda KT} - 1$$

$$E_\lambda = \frac{8\pi KT}{\lambda^4} d\lambda$$

Rayleigh Jeans Law

Based on assumptions, Planck's Energy equations successfully explained the **entire spectrum of black body**.

At long wavelength, Planck's equation reduces to **Rayleigh Jeans equation**.

At shorter wavelength, it predicts an exponential decrease in intensity with decreasing wavelength.

Einstein----- Confirmation of Quanta

After 5 years, Einstein confirmed that these **packets of energy are photons.**

Light consists of Photon, each with a particular amount of energy, called a **quanta of energy.**

$$E=mc^2$$

$$E=mc.c$$

$$E=pc \rightarrow \text{particle}$$

$$pc = \frac{hc}{\lambda}$$

$$p = \frac{h}{\lambda}$$

$$E = \frac{hc}{\lambda}$$

Wave

$$E = \frac{hc}{\lambda}$$

$$P_{\text{particle}} = P_{\text{Wave}}$$

$$mv = \frac{h}{\lambda}$$

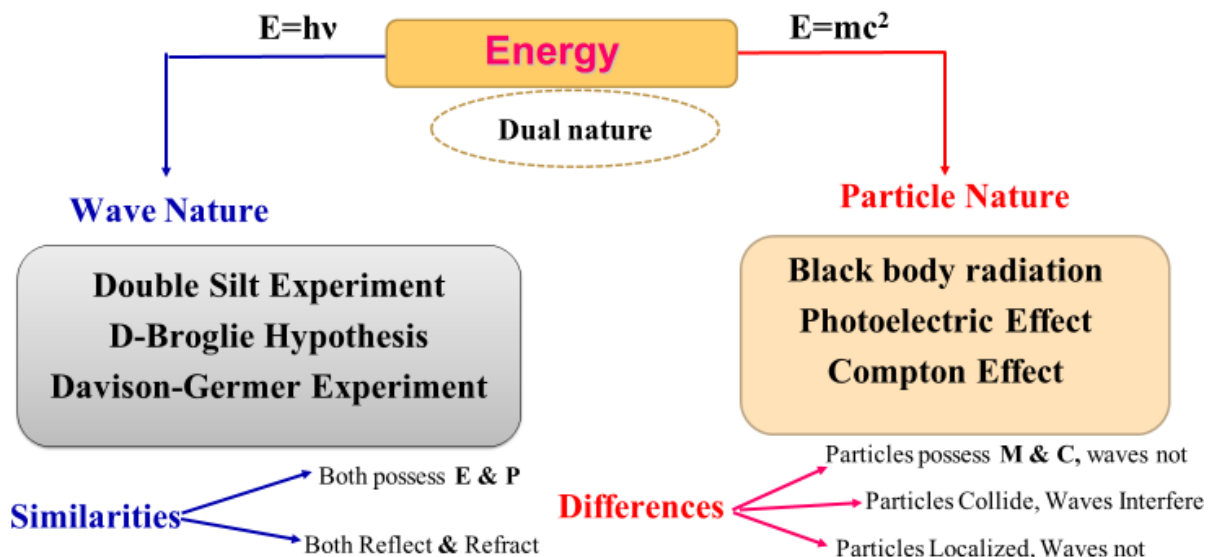
$$\lambda = \frac{h}{mv}$$

d-Broglie hypothesis

$$\lambda = \frac{h}{mv} = \frac{h}{p} = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2meV}}$$

$$E=eV$$

Origins of Quantum Mechanics



Real Time Examples

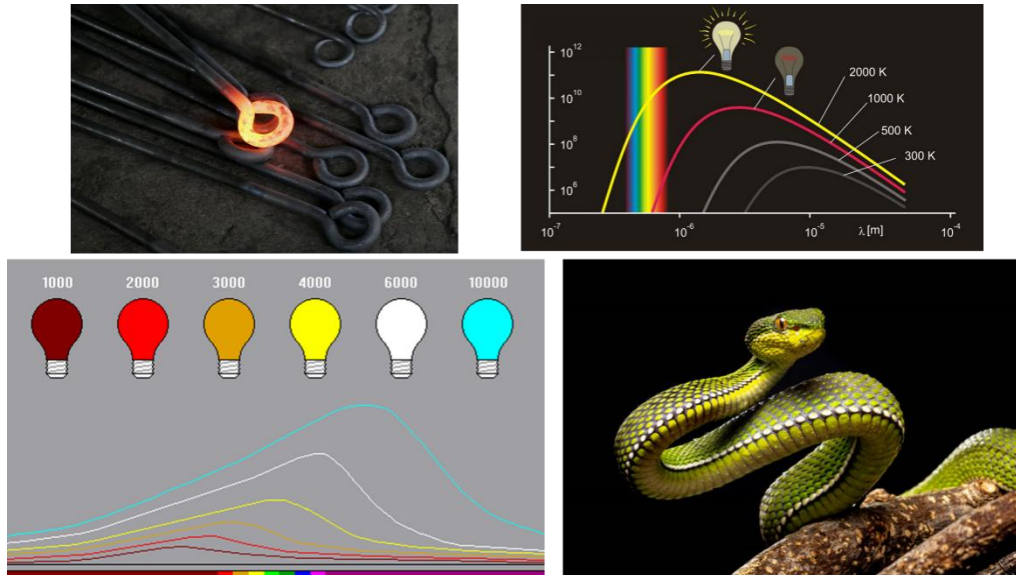
Stefan's Law or Stefan's Boltzmann's Law

An ear thermometer measures a patient's temperature by detecting the intensity of Infrared radiation leaving the eardrum.



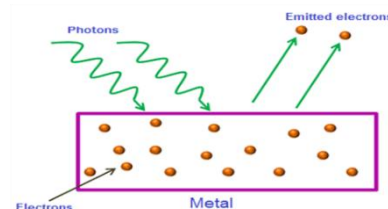
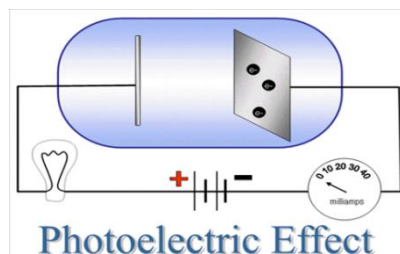
The energy radiated by a blackbody radiator per second per unit area is directly proportional to the fourth power of the absolute temperature.

Real Time Examples



The Photoelectric Effect

A beam of light hitting a metal surface can cause electrons to be ejected from the surface.



Influence of radiation such as γ rays, X-rays, ultraviolet and even visible light

- ✓ Effect of intensity of incident radiation on photo electric current
- ✓ Effect of potential difference on the photoelectric current
- ✓ Effect of frequency of incident radiation on stopping potential



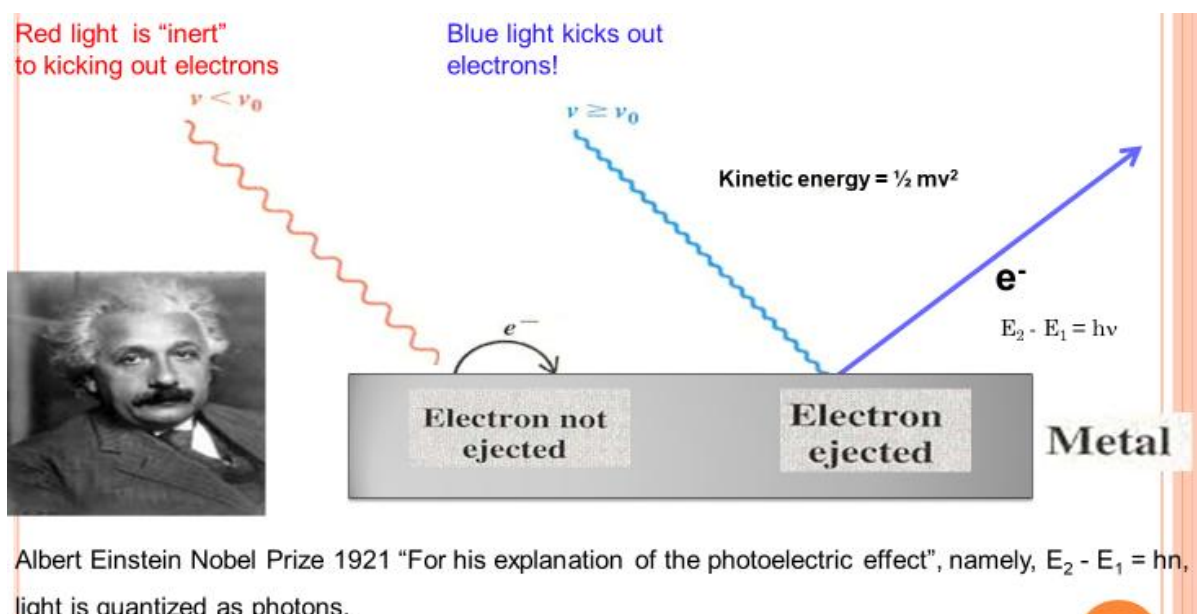
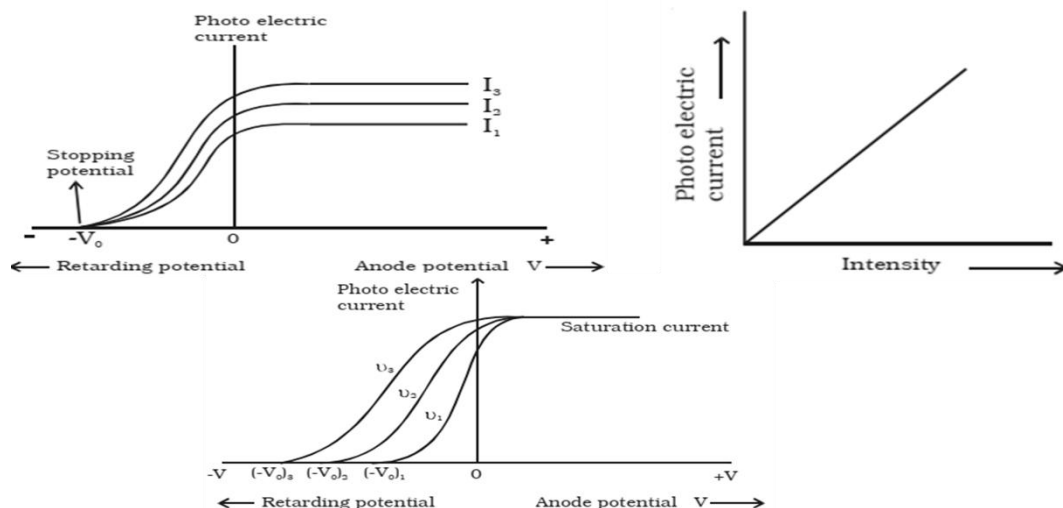
Stay Ahead

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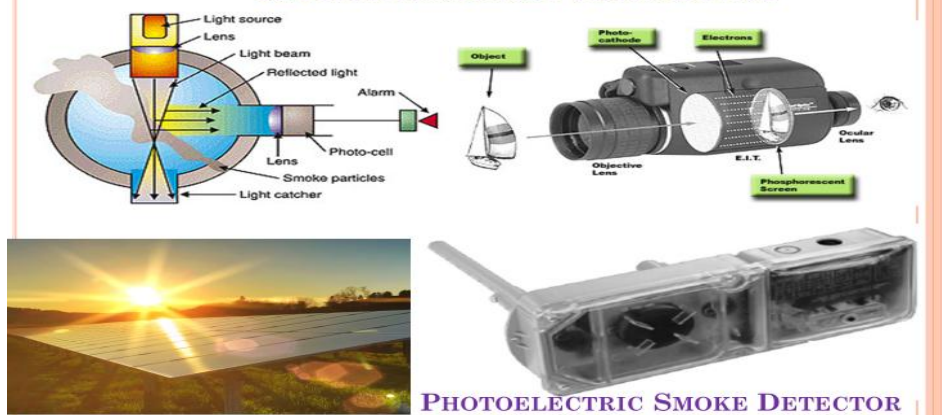


Albert Einstein Nobel Prize 1921 "For his explanation of the photoelectric effect", namely, $E_2 - E_1 = h\nu$, light is quantized as photons.

Applications of photo electric cells

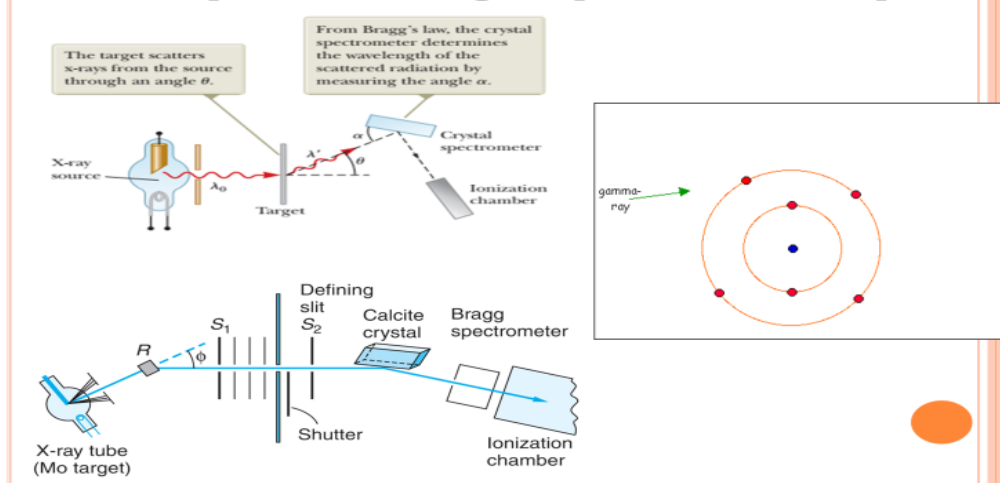
- (i) Photoelectric cells are used for reproducing sound in cinematography.
- (ii) They are used for controlling the temperature of furnaces.
- (iii) Photoelectric cells are used for automatic switching on and off the street lights.
- (iv) Photoelectric cells are used in the study of temperature and spectra of stars.
- (v) Photoelectric cells are also used in obtaining electrical energy from sunlight during space travel.
- (vi) These cells are used in instruments measuring light illumination.
- (vii) These cells are used in opening and closing of door automatically.
- (viii) Photoelectric cells are used in burglar alarm and fire alarm.

APPLICATIONS: NIGHT VISION DEVICE



PHOTOELECTRIC SMOKE DETECTOR

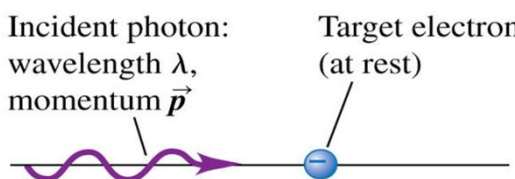
Compton scattering : Experimental Setup



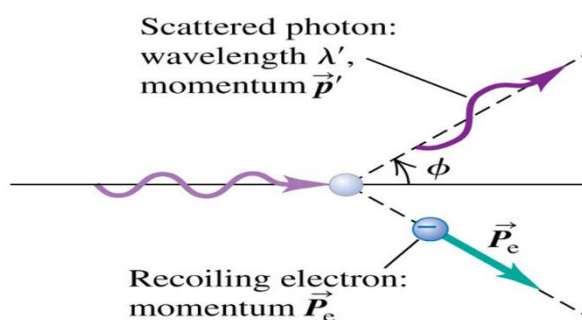
In the Compton experiment, x rays are scattered from electrons. The scattered x rays have a **longer wavelength than the incident x rays**, and the scattered wavelength depends on the scattering angle ϕ .

Explanation: When an incident photon collides with an electron, it transfers some of its energy to the electron. The scattered photon has less energy and a longer wavelength than the incident photon.

(a) Before collision: The target electron is at rest.



(b) After collision: The angle between the directions of the scattered photon and the incident photon is ϕ .



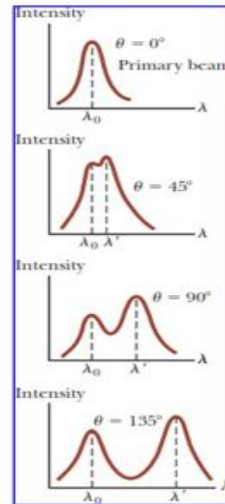
Compton shift equation

$$\lambda' - \lambda_0 = \frac{h}{m_e c} (1 - \cos \theta)$$

Compton wavelength

$$\lambda_C = \frac{h}{m_e c} = 0.00243 \text{ nm}$$

- The shift of the wavelength increased with scattering angle
- The shift of the wavelength is independent of the wavelength of the incident photon



Blackbody Radiation

- ❖ Photon Absorption and Emission
- ❖ Photoelectric Effect
- ❖ Photon Absorption
- ❖ Light (low energy) removes free electrons from the metal
- ❖ Proved the particle behavior of light
- ❖ Compton Effect
- ❖ Photon Scattering
- ❖ X-rays (high energy) removes bound electrons from carbon
- ❖ Proved the particle behavior of X-rays

The photons have no charge

They can interact with charged particles but not with themselves Photons are neutral and not affected by magnetic or electric fields.

De Broglie Waves

A moving body behaves in certain waves as though it has a wave nature

Momentum of a photon is given by $p = \frac{h\nu}{c} = \frac{h}{\lambda}$ since $\lambda\nu = c$

Photon Wavelength $\lambda = \frac{h}{p}$ (De Broglie suggested that this equation can be applied to material particles as well as photons)

Momentum of a particle with mass m and velocity v (Relativity Theory) $p = \gamma m v$

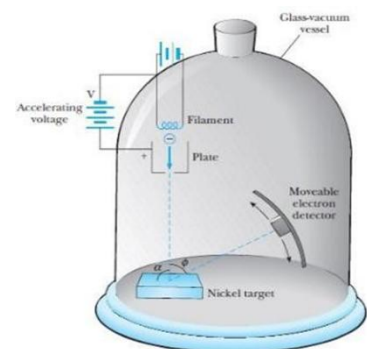
De Broglie Wavelength $\lambda = \frac{h}{\gamma m v}$ (Momentum greater $\rightarrow$ Wavelength shorter)

$$\text{Relativistic factor } \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$$

The wave and particle aspects of moving bodies can never be observed at the same time

In certain situations a moving body resembles a wave and in others it resembles a particle

Which type of property is more dominant depends on how its De Broglie wavelength compares with its dimensions and the dimensions of whatever it interacts with



(Davisson-Germer Experiment)
Scattering of electrons from a solid (Ni target)

Particle Diffraction

An experiment that confirms the existence of de Broglie waves

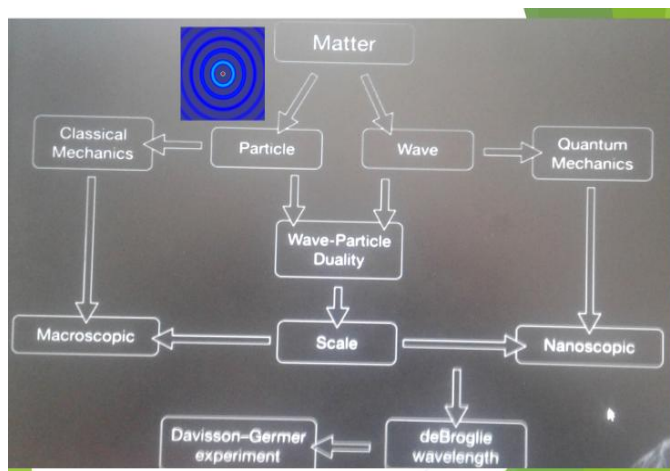
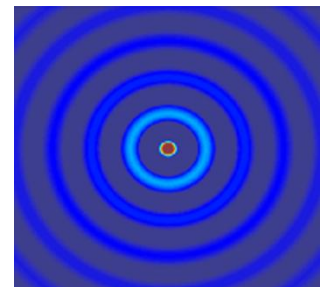
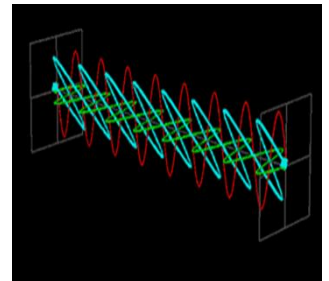
Diffraction is a wave effect with no analog in the behavior of Newtonian particles

In 1927, Davisson and Germer demonstrated that electron beams are diffracted when they are scattered by the regular atomic arrays of crystal

Electrons are not only particles but also waves

Matter waves

- A central part of the theory of quantum mechanics, being an example of wave-particle duality. All matter exhibits wave-like behavior.
- For example, a beam of electrons can be diffracted just like a beam of light or a water wave. In most cases, however, the wavelength is too small to have a practical impact on day-to-day activities.
- Hence in our day-to-day lives with objects the size of tennis balls and people, matter waves are not relevant.



Classical Mechanics and Quantum Mechanics

Classical Physics	Quantum Physics
A set of fundamental theory which describes matter at ordinary scale (macroscopic objects)	A modern theory which describes matter at the smallest scales (atomic and sub-atomic levels – microscopic objects)
Classical mechanics describes the behavior of objects in terms of two equations $E = \frac{p^2}{2m} + V(x)$ (Total energy is constant in the absence of external forces) $F = ma$ (Response of particles to the forces acting on them – Newton's 2 nd law)	All the properties of a system are expressed in terms of wavefunction that is obtained by solving the Schrödinger's wave equation $\psi(x) = A \sin \frac{n\pi}{L} x$ $\nabla^2 \psi + \frac{2m}{\hbar^2} (E - V) \psi = 0$ $\hbar \frac{d\psi}{dt} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi$

$$E = h\nu$$

frequency of radiation, sometimes written as f giving expression $E = hf$.
 Quantum energy of a photon.
 $h = \text{Planck's constant} = 6.626 \times 10^{-34} \text{ Joule-sec} = 4.136 \times 10^{-15} \text{ eV} \cdot \text{s}$

Electromagnetic radiation may be treated in classical or quantum-mechanical aspects

– Compare these two aspects

Max Planck's Theory

A hot body emits radiant energy not continuously but discontinuously in the form of small packets of energy called quantum (photon).